

V3 empirical study · Instruction

Two models · four underlyings · four volatility regimes · 1.5-year monthly calibration · option-implied Heston fami

What this report is

- Companion to the seven-model 10,000-path study. This report keeps only Heston and Heston–Merton, which are calibrated to listed calls (Fourier NLS). Rankings are within this pair; they are not compared here with return-based models.
- Companies are SPY, AAPL, MSFT, and AMZN. No new calibration or LSM. Every number is the stored cell from V3_1p5y_monthly_empirical_study.
- Return-based companion: V3_1p5y_monthly_empirical_study_return_based.pdf.
- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 2 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column says BEST.

Experimental design (held fixed for every model)

- Models: Heston (no jumps), Heston–Merton (Bates).
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

V3 empirical study · Instruction (continued)

Two models · four underlyings · four volatility regimes · 1.5-year monthly calibration · option-implied Heston fami

Shared market sample and filters (not model-specific)

- For each company $\times$ regime, one listed-call sample is drawn once and reused by all two models in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S , K , C ; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05 , valid bid-ask with relative spread $\leq 50\%$ when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: $(\kappa, \theta, \xi, \rho, v_0)$ are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \rightarrow r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t ; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud ($n_{\text{paths}} = 10000$, seed 42, $n_{\text{steps}} = \text{DTE}$, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE} = \text{mean} |C_{\text{model}} - C_{\text{mkt}}|$ (dollars). $\text{Bias \$} = \text{mean}(C_{\text{model}} - C_{\text{mkt}})$ (dollars). $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before expiry.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all two models in this report)

Regime	Window	SPY n	AAPL n	MSFT n	AMZN n
2008-2009 Crisis	2008-08-01 → 2009-07-31	50	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-31	50	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-30	50	51	50	51
2019-2020 COVID	2019-09-01 → 2020-08-31	50	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7-60	S/K-1 ≤10%	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655
AMZN	2008-2009	465	465	465	442	442
AMZN	2013-2014	5715	5692	5692	5598	5598
AMZN	2018-2019	28809	28808	28808	28346	28346
AMZN	2019-2020	26766	26760	26760	26323	26323

*Input is the processed call panel restricted to (regime end – 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with $\text{trading_date} \leq t$. LSM prices the Monday contract using the latest calibration row with date $\leq$ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All two models in this report still share that exact sample.

Heston / Heston-Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity $\times$ moneyness strata). GBM / Merton / GARCH / GARCH-Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	32.75	1.0960	+0.5803	+20.80	26.9%	2
Heston-Merton	31.60	1.0435	+0.5375	+19.11	21.4%	1

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	52.46	1.0201	+0.9451	+42.44	26.9%	1
Heston-Merton	67.16	1.2748	+1.2103	+53.41	24.5%	2

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	73.93	2.3623	+2.1183	+54.85	22.3%	1
Heston-Merton	84.01	2.7641	+2.5349	+64.41	17.0%	2

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	65.44	3.7704	+2.6179	+50.30	26.4%	1
Heston-Merton	66.64	3.5171	+2.3874	+47.68	15.7%	2

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 5 · Table block · Company summary · two models × four regimes

Each number is the arithmetic mean of AAPL, MSFT, AMZN. Same columns as the SPY page. Bold green row = lowest mean RMSE%. 1.5-year lookahead, rolling=monthly, n_paths=

Table Co-1 Company summary · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	29.18	0.9812	+0.6597	+18.06	24.1%	1
Heston-Merton	29.62	0.9867	+0.6635	+18.32	17.9%	2

Table Co-2 Company summary · 2013-2014 Normal
2014-01-01 → 2014-12-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	36.31	2.0299	+1.8887	+26.42	21.7%	2
Heston-Merton	32.05	1.7232	+1.5304	+20.94	14.7%	1

Table Co-3 Company summary · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · mean of AAPL, MSFT, AMZN · $\bar{n} = 51$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	40.44	7.4928	+6.8740	+31.55	21.1%	2
Heston-Merton	37.27	6.7082	+5.8681	+26.30	16.8%	1

Table Co-4 Company summary · 2019-2020 COVID
2019-09-01 → 2020-08-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	42.95	11.7492	+7.8777	+31.35	22.9%	2
Heston-Merton	40.20	11.3578	+7.0705	+25.96	16.7%	1

RMSE%, MAE, Bias \$, Bias%, and early-exercise fraction are each averaged across AAPL, MSFT, AMZN. This is not a pooled re-pricing of a combined contract sample. Ranking uses the mean RMS

Page 6 · Table block · AAPL · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	31.71	1.5996	+1.0721	+21.09	22.7%	1
Heston-Merton	32.30	1.6138	+1.0704	+21.37	18.5%	2

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	37.85	2.5745	+2.5702	+29.62	20.8%	2
Heston-Merton	31.91	1.9670	+1.9420	+21.53	15.2%	1

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	45.14	2.3820	+2.2976	+38.59	21.0%	2
Heston-Merton	41.69	2.0875	+1.9310	+32.87	18.2%	1

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	42.24	3.7296	+2.9444	+33.15	23.3%	2
Heston-Merton	40.88	3.4396	+2.5150	+30.03	16.7%	1

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 7 · Table block · MSFT · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	26.51	0.2560	+0.0897	+12.61	24.7%	1
Heston-Merton	26.84	0.2552	+0.0910	+12.75	18.6%	2

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	31.79	0.1889	+0.1565	+21.39	22.6%	2
Heston-Merton	26.54	0.1648	+0.1181	+16.70	12.2%	1

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	37.47	0.9577	+0.8192	+26.22	21.8%	2
Heston-Merton	33.23	0.7984	+0.6200	+19.90	16.2%	1

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	40.67	2.0782	+1.5079	+30.43	23.8%	2
Heston-Merton	35.19	1.7711	+1.0681	+20.23	16.0%	1

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 8 · Table block · AMZN · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	29.31	1.0880	+0.8173	+20.47	25.0%	1
Heston-Merton	29.73	1.0912	+0.8291	+20.84	16.5%	2

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	39.29	3.3265	+2.9394	+28.25	21.8%	2
Heston-Merton	37.71	3.0378	+2.5310	+24.58	16.6%	1

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	38.70	19.1386	+17.5052	+29.83	20.5%	2
Heston-Merton	36.89	17.2386	+15.0534	+26.13	16.0%	1

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	45.95	29.4399	+19.1809	+30.46	21.6%	2
Heston-Merton	44.52	28.8627	+17.6284	+27.61	17.3%	1

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	Heston-Merton	41.68	36.04	10
2	Heston	41.95	39.00	6

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
Heston	3	1	1	1	6
Heston-Merton	1	3	3	3	10

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
Heston	30.07	40.35	48.81	48.58	41.95
Heston-Merton	30.12	40.83	48.96	46.81	41.68

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
Heston	56.15	58.95	3
Heston-Merton	62.35	66.90	1

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
Heston	39.23	40.05	1
Heston-Merton	36.69	36.59	3

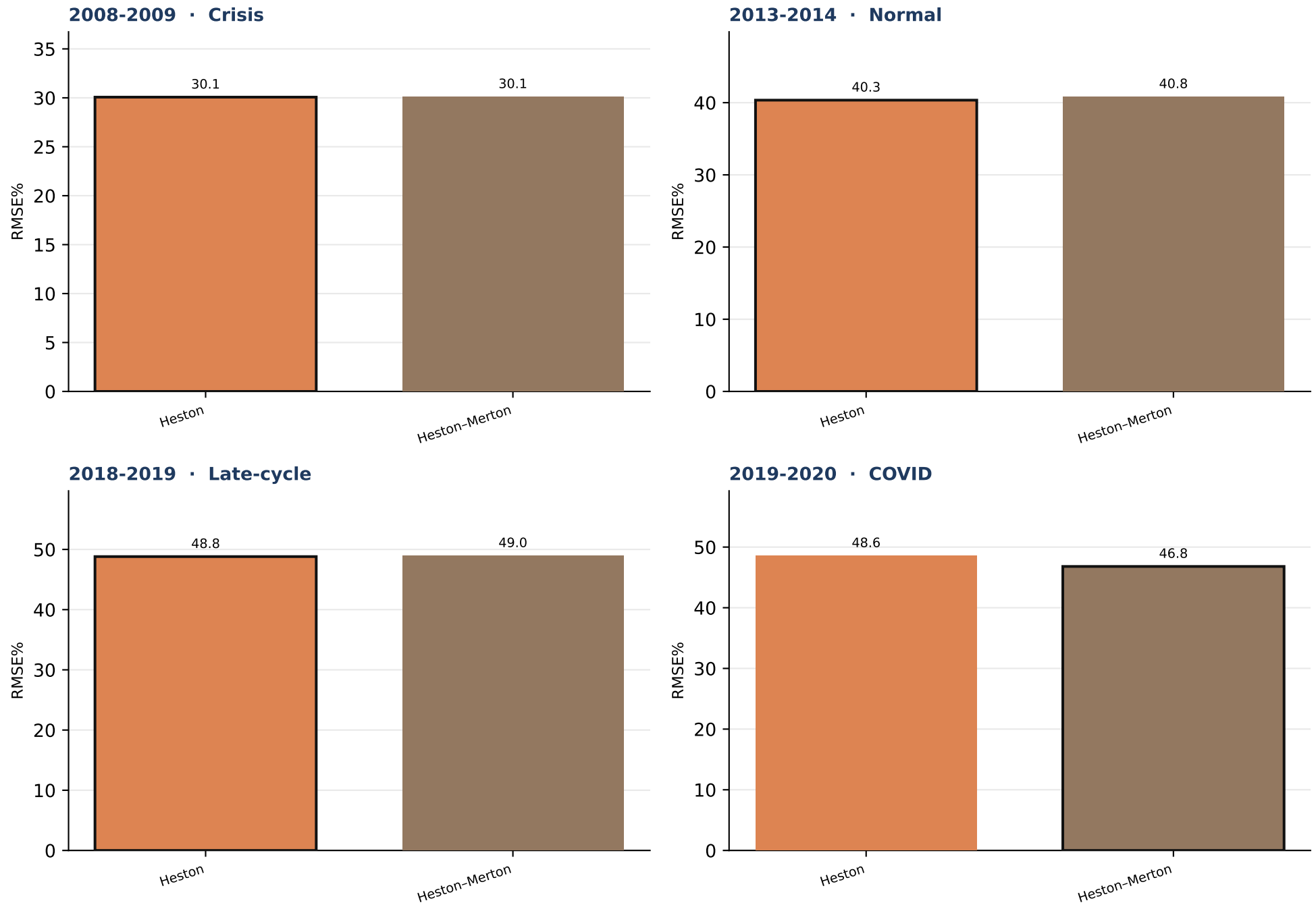
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
Heston	34.11	34.63	1
Heston-Merton	30.45	30.03	3

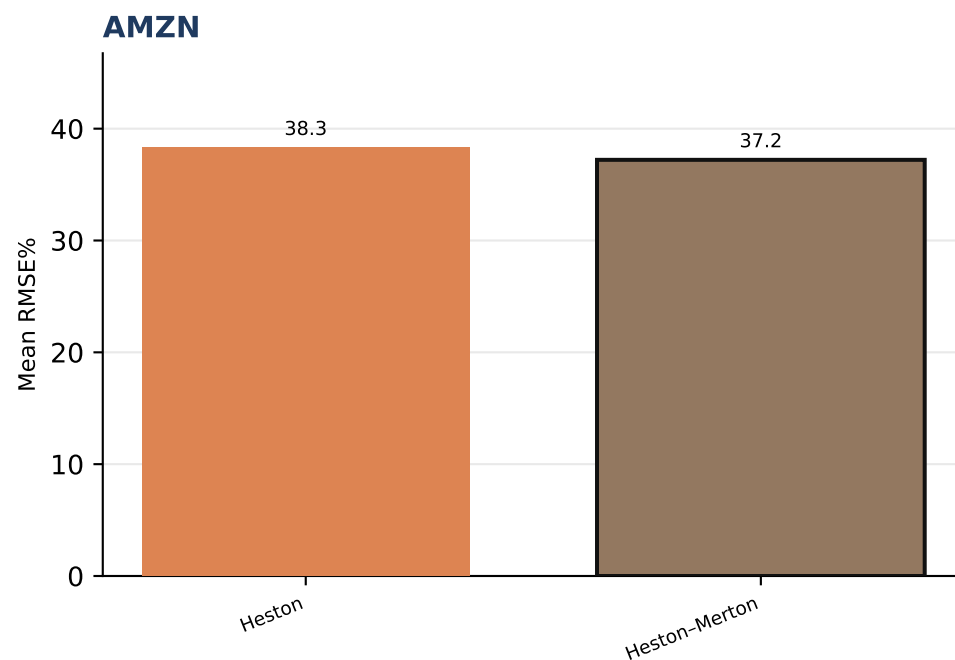
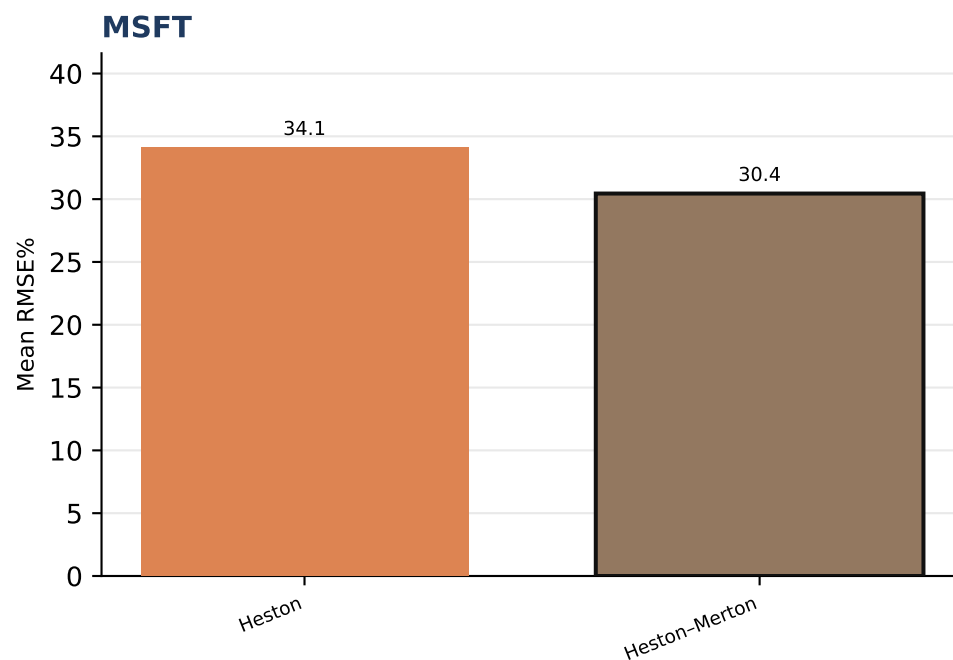
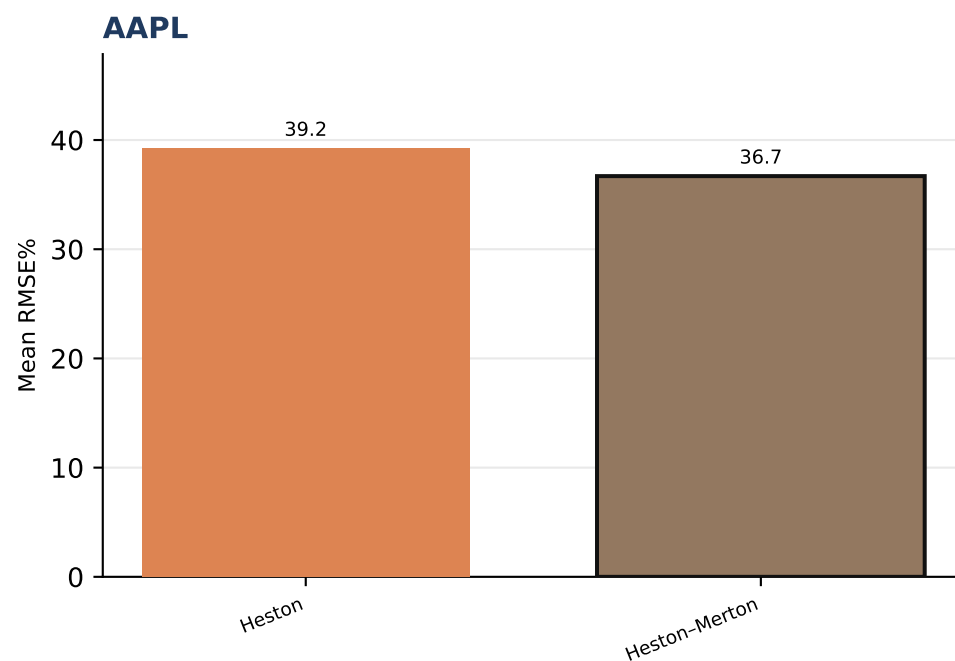
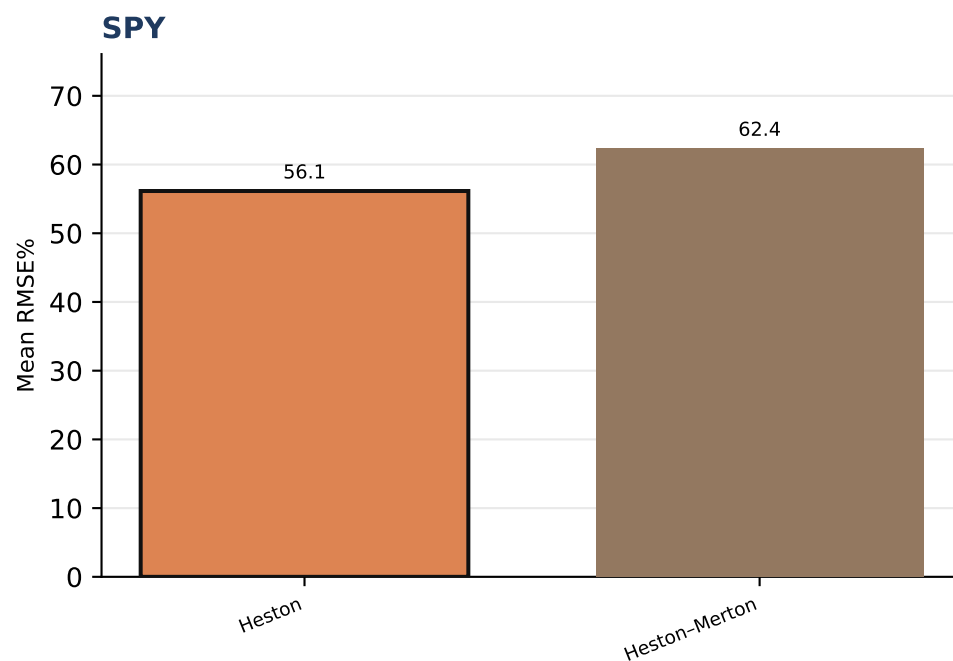
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
Heston	38.31	39.00	1
Heston-Merton	37.21	37.30	3

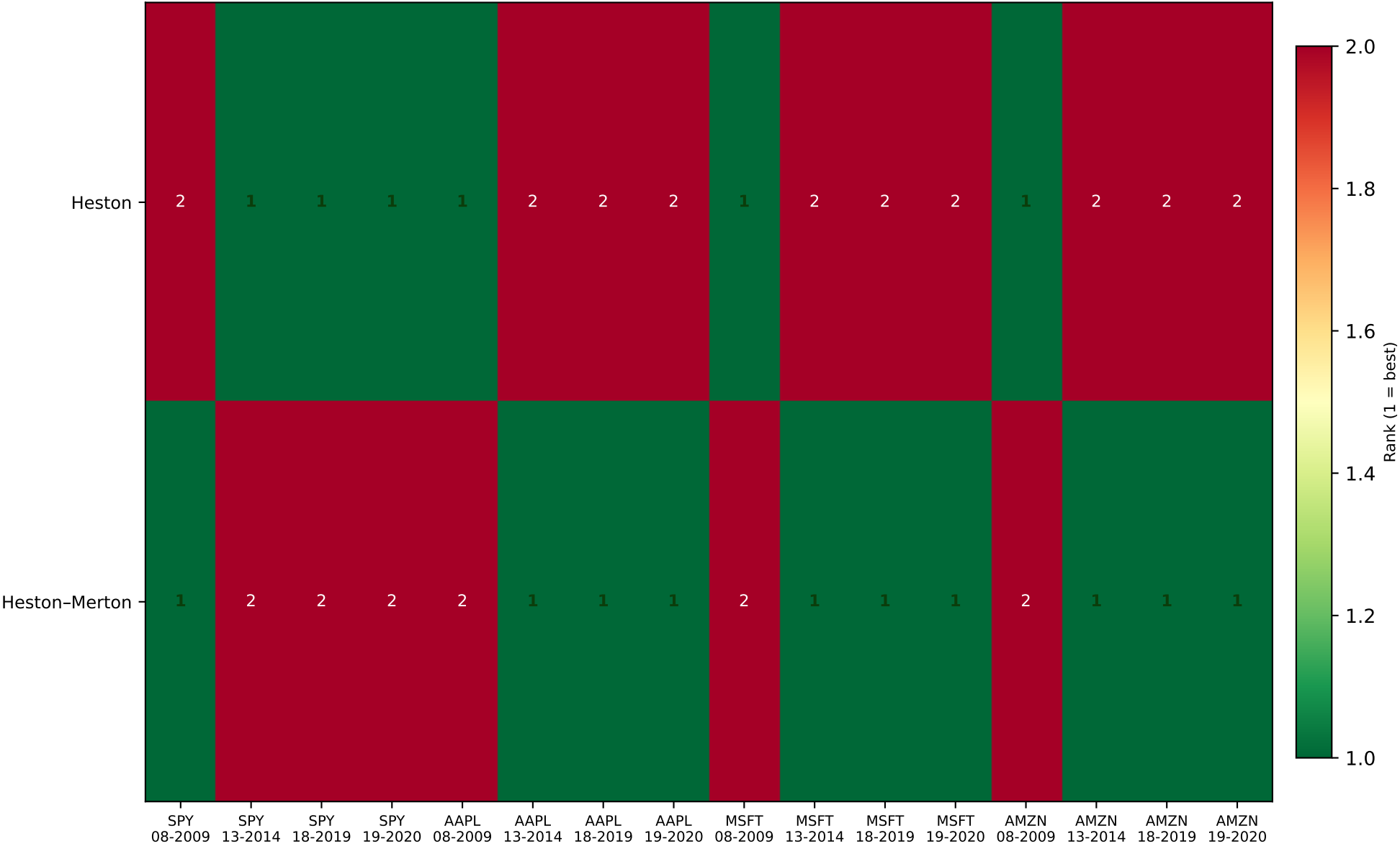
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



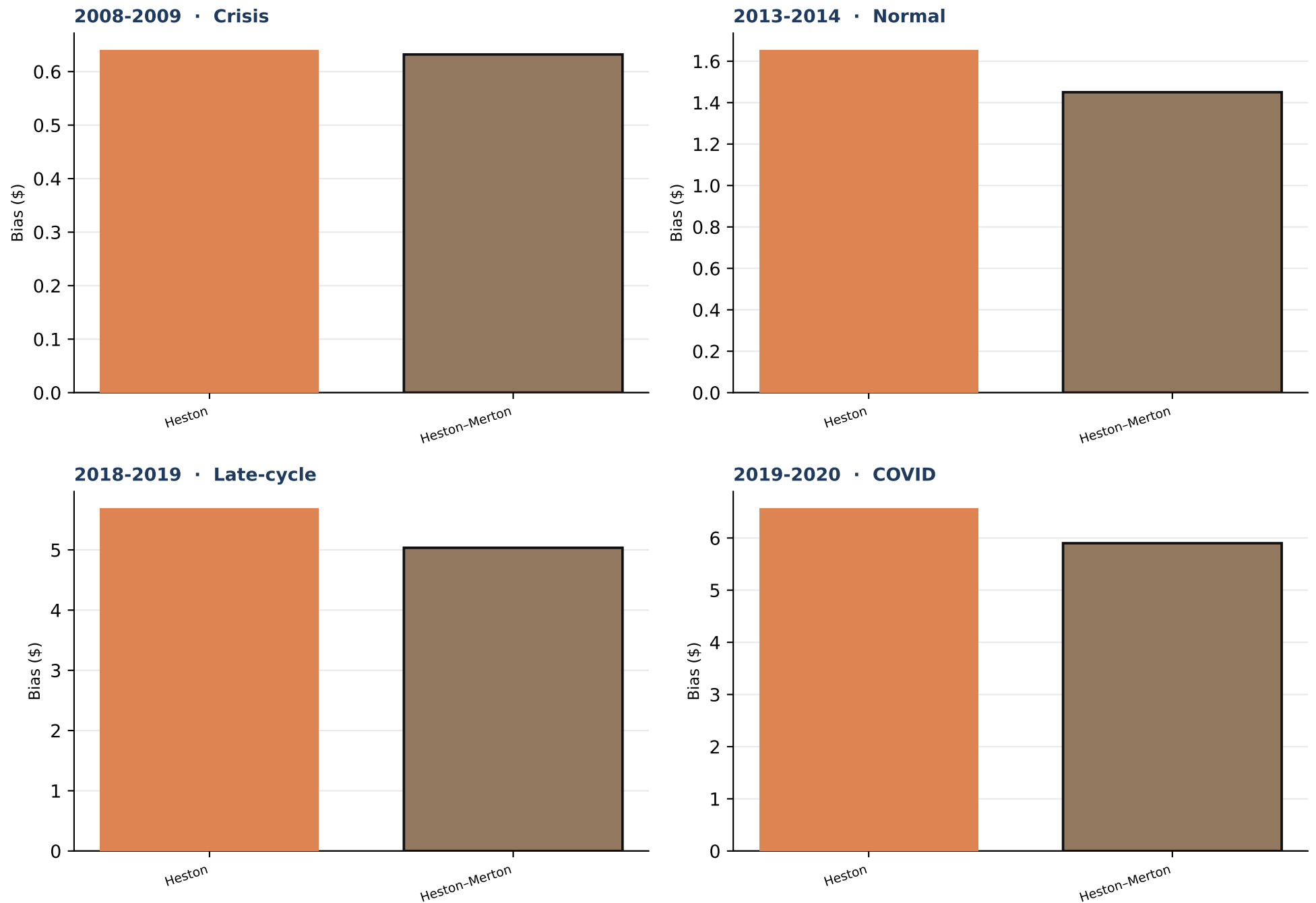
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, Heston-Merton is the best overall model (mean RMSE% = 41.68; 10 of 16 cells).
- Second overall is Heston (mean RMSE% = 41.95; 6 cells).
- The ranking is not stable. Best-in-cell models across the 16 tables: Heston, Heston-Merton.
- SPY: lowest mean RMSE% is Heston (56.15); wins 3 of 4 regimes.
- AAPL: lowest mean RMSE% is Heston-Merton (36.69); wins 3 of 4 regimes.
- MSFT: lowest mean RMSE% is Heston-Merton (30.45); wins 3 of 4 regimes.
- AMZN: lowest mean RMSE% is Heston-Merton (37.21); wins 3 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is Heston (30.07).
- 2013-2014 (Normal): best mean RMSE% is Heston (40.35).
- 2018-2019 (Late-cycle): best mean RMSE% is Heston (48.81).
- 2019-2020 (COVID): best mean RMSE% is Heston-Merton (46.81).
- Jump models as a group have lower mean RMSE% (41.68) than the no-jump models in this report (41.95).
- For the overall winner (Heston-Merton), crisis RMSE% is 30.12 vs 40.83 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year monthly calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all two models in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE, Bias \$, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).
- Early-exercise fractions are a model implication, not an accuracy score; American calls on these